

Log-Location-Scale Survival Models for Heart Failure Data: **A Comparative Study**

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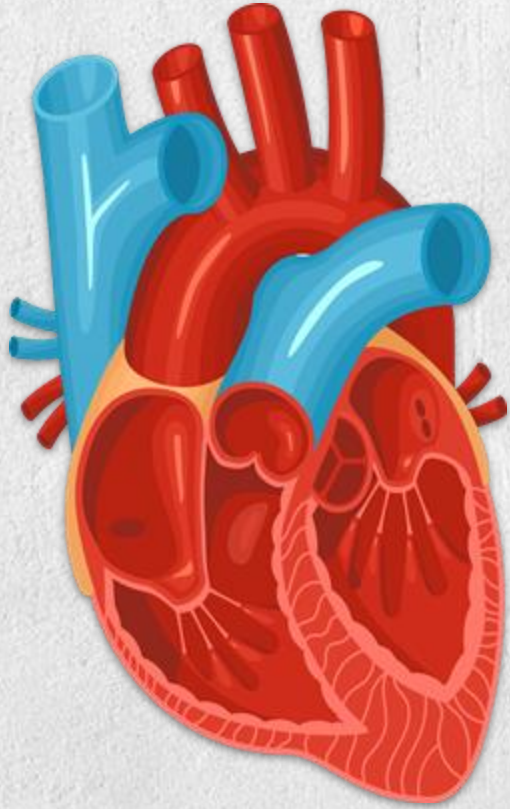
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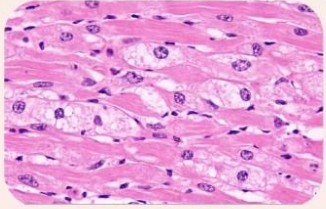


Introduction

01

What is Heart Failure?

Heart Failure (HF) is a chronic and progressive clinical syndrome. It occurs when the heart muscle is structurally or functionally impaired, making it unable to pump enough blood to meet the body's metabolic demands for oxygen and nutrients.



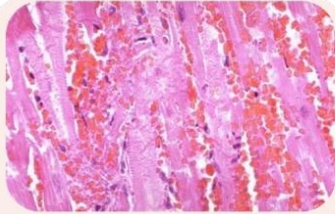
Phase 1: The Struggling Heart

- **Serum Creatinine**

Normal or slightly rising.

- **Ejection Fraction**

Drops suddenly



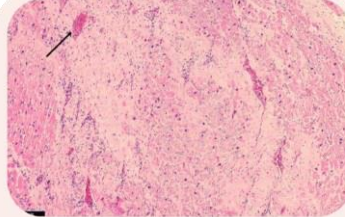
Phase 2: Cell Death

- **Serum Creatinine**

Rises quickly and sharply.

- **Ejection Fraction**

Drops dangerously low.



Phase 3: Healing & Scarring

- **Serum Creatinine**

Settles at a stable.

- **Ejection Fraction**

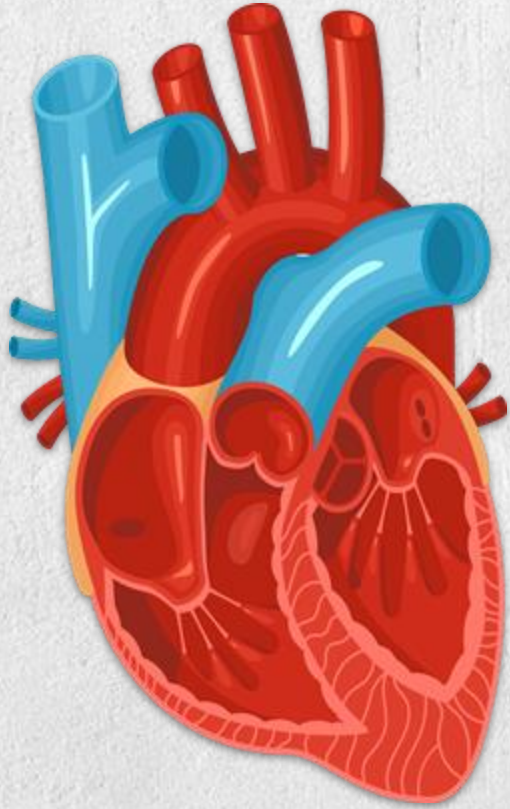
Stays permanently low



Core Elements of the Study

- Study Goal: To improve the modeling of complex and heterogenous HF survival data.
- The Problem: Different statistical models make different assumptions, which drastically changes how covariate effects (risk factors) are interpreted. So Instead of relying on a single model, this study compares multiple parametric models (Weibull, Log-normal, Log-logistic) using real HF data.
- Clinical Significance: Finding the best-fitting model helps correctly identify variables that speed up or slow down disease progression, leading to optimized patient treatment and valid statistical inference.





Literature Review

02

2020

Feature Importance in ML

Chicco - Jurman
Serum creatinine and ejection fraction alone predict HF survival effectively

2022

Exponentiated Odd LLS

Rodrigues
Introduced generalized log-transformations for demographic data.

2024

Bayesian Parametric AFT

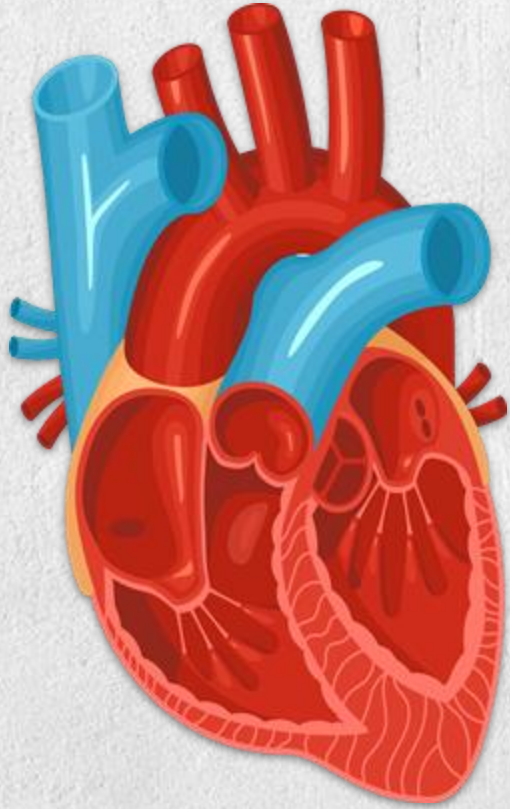
Ashine
Log-normal AFT provides the best fit via INLA.

2025

Parametric AFT Evaluation

LI
Log-normal framework consistently produces top results.

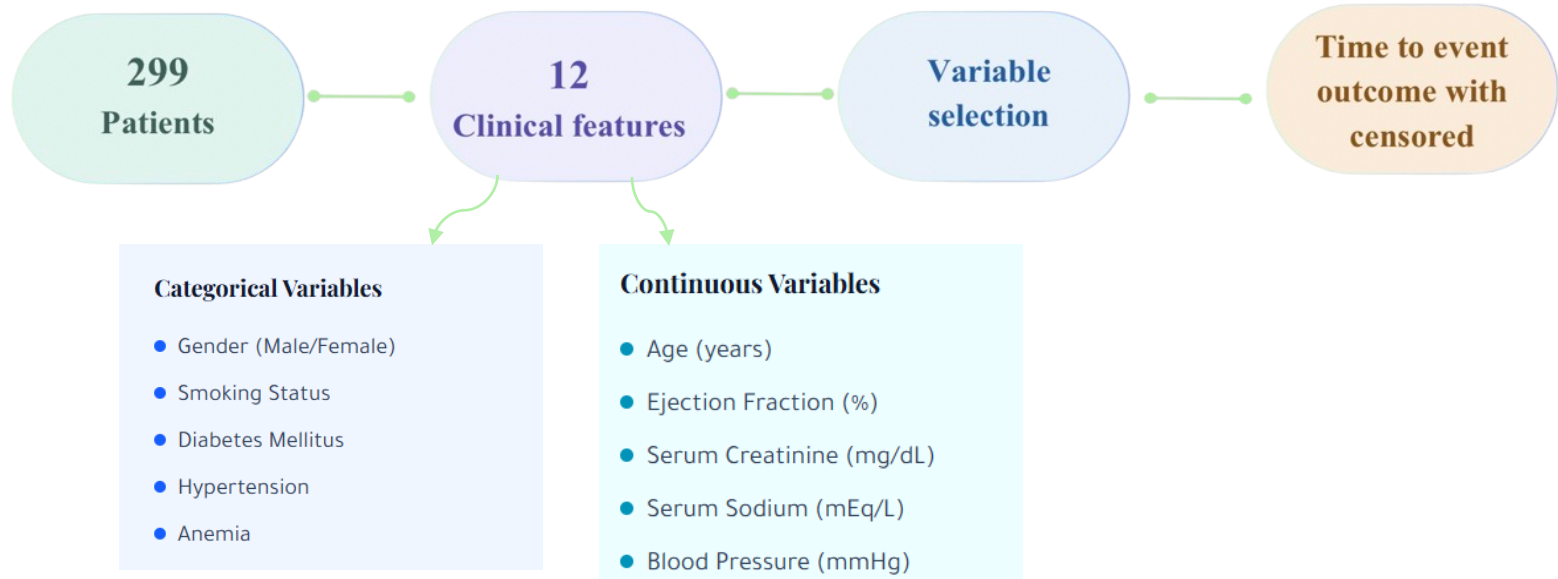




Methodology

03

Data Description



LLS Framework and AFT Regression

Survival time is modelled using the Accelerated Failure Time (AFT) approach within the log-location-scale (LLS) framework:

$$\log(T_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + \sigma \varepsilon_i, i = 1, 2, \dots, n$$

where $\boldsymbol{\beta}$ represents regression coefficients, σ is the scale parameter, and ε_i follows a specific distribution.

Weibull



$$\ell(\boldsymbol{\beta}, \sigma) = \sum_{i=1}^n [\delta_i (-\log \sigma - \log t_i + z_i) - \exp(z_i)]$$

Log-normal



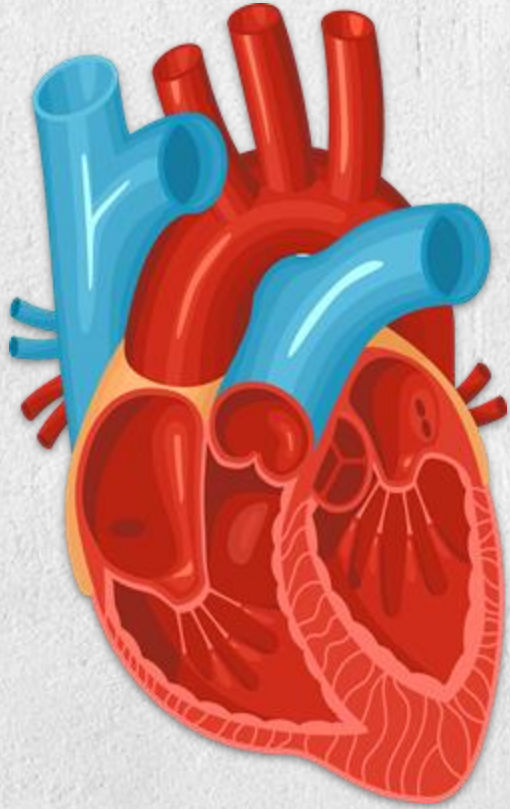
$$\ell(\boldsymbol{\beta}, \sigma) = \sum_{i=1}^n \left[\delta_i \left(-\log \sigma - \log t_i - \frac{1}{2} \log(2\pi) - \frac{1}{2} z_i^2 \right) + (1 - \delta_i) \log(1 - \Phi(z_i)) \right]$$

Log-logistic



$$\ell(\boldsymbol{\beta}, \sigma) = \sum_{i=1}^n [\delta_i (-\log \sigma - \log t_i + z_i - 2 \log(1 + \exp(z_i))) - (1 - \delta_i) \log(1 + \exp(z_i))]$$





Results

04

Model Selection Criteria

- Variable selection: **Backward selection**

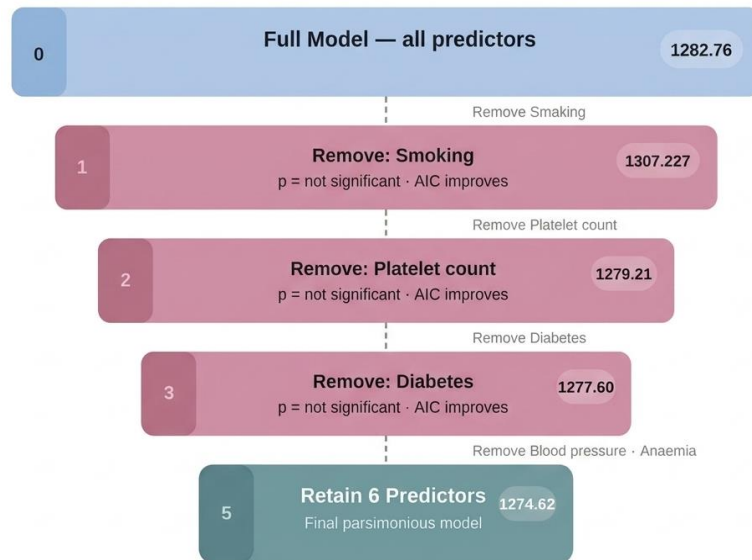


Figure 1: AIC-guided backward elimination procedure for the Weibull AFT survival model.



Model Selection Criteria

- **Model comparison: AIC, BIC, CAIC and Log Likelihood**

Table1: Parametric Survival Model Comparison Statistics

MODEL	LOGLIK	AIC	BIC	CAIC
WEIBULL	-630.812	1277.624	1307.227	1315.227
LOG-NORMAL	-633.279	1282.557	1312.161	1320.161
LOG-LOGISTIC	-632.179	1280.359	1309.962	1317.962

- The Weibull model has the lowest AIC, BIC, Log Likelihood, and CAIC, confirming it as the best-fitting model.



Model Selection Criteria

- Regression Model : **Weibull**

Table2: Weibull Regression Model Coefficients

Variable	coefficient	Std.Error	p-value
Age	-0.04743	0.00971	0.001
Ejection Fraction	0.05043	0.01132	0.001
Serum Creatinine	-0.31606	0.07277	0.001



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- The Weibull regression identifies Age, Ejection Fraction, and Serum Creatinine as the most significant predictors.



Model Selection Criteria

- Regression Model : **Weibull**

Three clinical predictors explain survival time.

Age



$$\beta = -0.04743$$

Ejection fraction



$$\beta = 0.05043$$

Serum creatinine



$$\beta = -0.31606$$



Kaplan-Meier Estimation

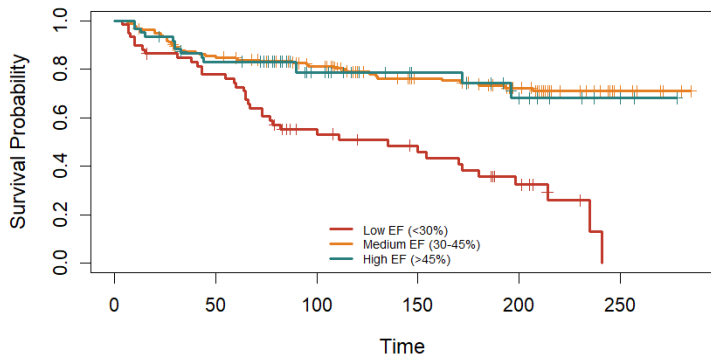


Figure 2: Kaplan-Meier by Ejection Fraction

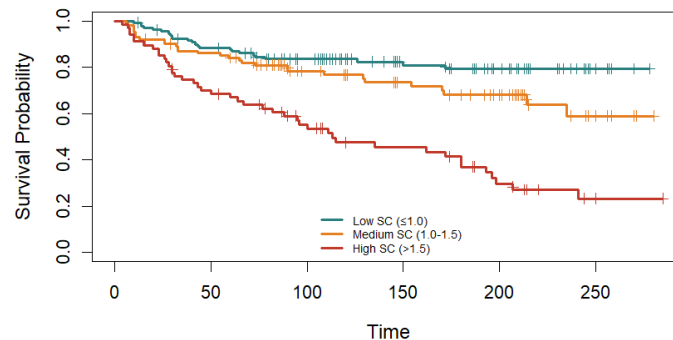


Figure 3: Kaplan-Meier by Serum Creatinine

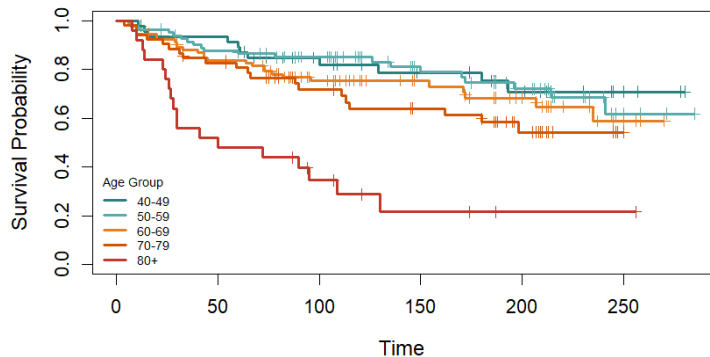


Figure 4: Kaplan-Meier Curves by Age Group



Model Diagnostics and Validation

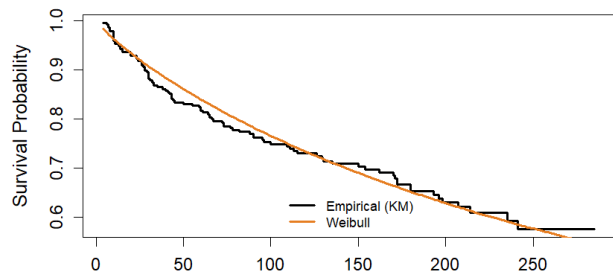


Figure 5: Kaplan-Meier Weibull fitted — empirical

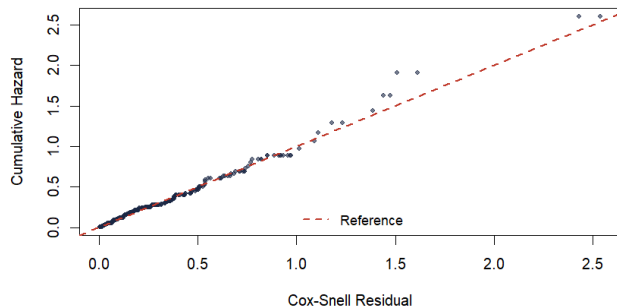
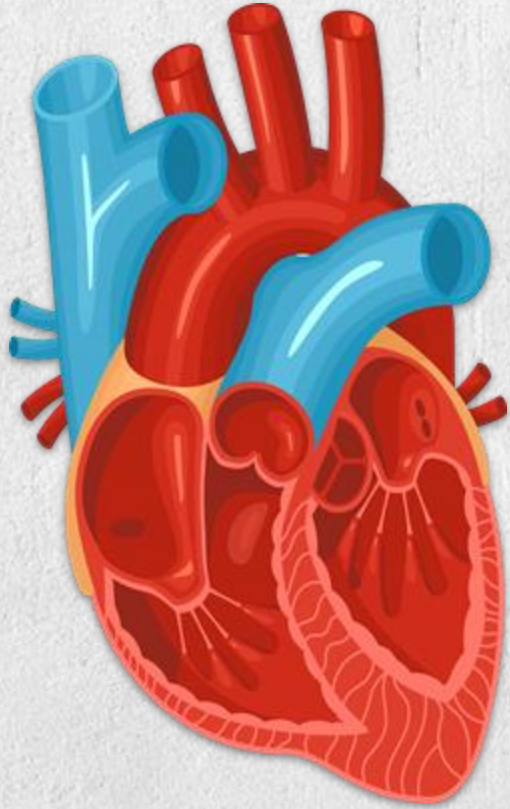


Figure 6: Cox-Snell — Weibull

Table 3: Variance Inflation Factors (VIF) for the Generalized Predictors

Variable	Variance Inflation Factor (VIF)
Age	1.043
Anemia	1.013
Ejection Fraction	1.065
Serum Creatinine	1.012
Serum Sodium	1.041
Sex	1.046





Conclusion and Future Work

05

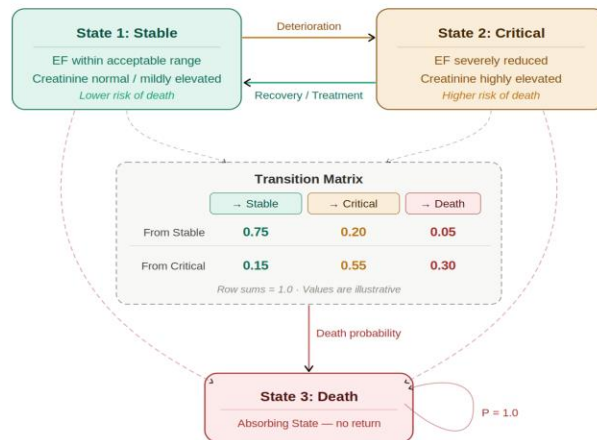
Conclusion

In conclusion, the model shifts from a larger set of variables to a more refined structure based on three key predictors. The final model highlights the importance of age, cardiac function, and renal function in explaining survival outcomes, while maintaining a good balance between model simplicity and predictive performance. off between accuracy and model complexity.

Future Work

Another important extension is to specify a multi-state survival model. The model can be extended to include intermediate health states such as hospitalization or disease progression rather than the simple transition between survival and death. This offers a clearer view of how patient conditions evolve and a better understanding of the natural history of HF over time.

Markov Model — Patient Health State Transitions





We recommend that decision-makers focus on the three variables affecting the disease, as well as early warning systems for kidney and heart failure, and that a treatment strategy be provided. We also recommend that hospitals implement the Whipple model.



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Thanks

Do you have any questions?



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